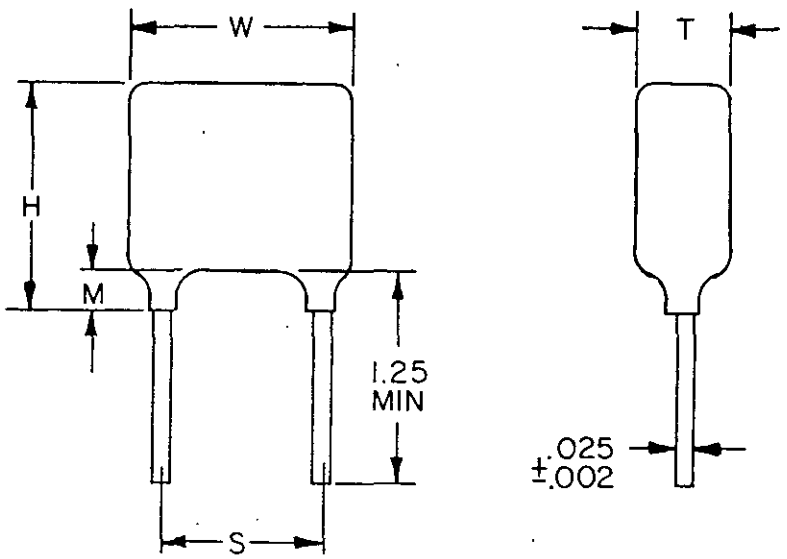
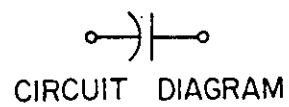
MILITARY SPECIFICATION SHEET

Capacitors, Ceramic, Multilayer, High Voltage, 5000 V DC, Established Reliability, Style HVR05

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-49467.





Case code	Dimensions				
	W max	H max	T max	M max	S ±.030
E	.570	.500	.270	.060	.475
F	.670	.600	.270	.060	.575
G	.770	.720	.270	.060	.675
J	1.300	.600	.270	.060	1.175
K	1.500	.720	.270	.060	1.375

Inches	mm	Inches	mm
.002	.05	.670	17.02
.025	.64	.675	17.14
.030	.76	.720	18.29
.060	1.52	.770	19.56
.270	6.86	1.175	29.84
.475	12.06	1.250	31.75
.500	12.70	1.300	33.02
.570	14.48	1.375	34.92
.575	14.60	1.500	38.10
.600	15.24		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Capacitor style HVR05.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case type: Conformally coated, molded, or encapsulated.

Lead material: In accordance with MIL-STD-1276.

Capacitance value: See table I.

Capacitance tolerance: J = ±5%, K = ±10%, M = ±20%. See table I.

Operating temperature range: -55°C to +125°C.

Voltage rating: 5000 V dc.

Failure rate level: M (1.0%), P (0.1%), R (.01%), or S (.001%).

Dielectric types: BP, BR, and BZ in accordance with MIL-C-49467.

Temperature coefficient: 0 ±30 ppm/°C (BP), ±15% (BR, BZ).

Voltage-temperature limits: In accordance with MIL-C-49467.

Symbol	Capacitance change with reference to +25°C	
	Steps A through D of voltage-temperature limit cycle table of MIL-C-49467. Bias = 0 volt.	Steps E through G of voltage-temperature limit cycle table of MIL-C-49467.
BP	0 ±30 ppm/°C	100% rated voltage: 0 ±30 ppm/°C
BR	±15%	100% rated voltage: +15, -40% 40% rated voltage: +15, -25% 60% rated voltage: +15, -40%
BZ	±15%	

Dissipation factor (25°C): 2.5% maximum (BR and BZ characteristics) or 0.1% maximum (BP characteristic), measured under the same conditions as capacitance.

Resistance to soldering heat: ΔC: -1% to +2% for BP characteristic.
-1% to +6% for BR and BZ characteristics.

Marking: In accordance with MIL-C-49467.

APPLICATION NOTES:

Additional encapsulation is necessary in applications where the possibility of a voltage breakdown between the leads of the capacitor, or the capacitor to another potential, could occur.

Heat sinks on each lead or adequate preheating is required when these capacitors are installed in or removed from circuits by soldering iron.

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TABLE I. Capacitor characteristics.

PIN 1/	Capacitance	Capacitance tolerance	Characteristic	Case code
M49467P05270__	27 pF	J, K	BP	E
M49467P05330__	33	J, K	BP	E
M49467P05390__	39	J, K	BP	E
M49467P05470__	47	J, K	BP	E
M49467P05560__	56	J, K	BP	E
M49467P05680__	68	J, K	BP	E
M49467P05820__	82	J, K	BP	E
M49467P05101__	100	J, K	BP	E
M49467_05101__	100	K, M	BR, BZ	E
M49467P05121__	120	J, K	BP	E
M49467_05121__	120	K, M	BR, BZ	E
M49467P05151__	150	J, K	BP	E
M49467_05151__	150	K, M	BR, BZ	E
M49467P05181__	180	J, K	BP	E
M49467_05181__	180	K, M	BR, BZ	E
M49467P05221__	220	J, K	BP	E
M49467_05221__	220	K, M	BR, BZ	E
M49467P05271__	270	J, K	BP	E
M49467_05271__	270	K, M	BR, BZ	E
M49467P05331__	330	J, K	BP	E
M49467_05331__	330	K, M	BR, BZ	E
M49467P05391__	390	J, K	BP	E
M49467_05391__	390	K, M	BR, BZ	E
M49467P05471__	470	J, K	BP	E
M49467_05471__	470	K, M	BR, BZ	E
M49467P05561__	560	J, K	BP	E
M49467_05561__	560	K, M	BR, BZ	E
M49467P05681__	680	J, K	BP	E
M49467_05681__	680	K, M	BR, BZ	E
M49467P05821__	820	J, K	BP	E
M49467_05821__	820	K, M	BR, BZ	E
M49467P05102__	1000	J, K	BP	E
M49467_05102__	1000	K, M	BR, BZ	E
M49467P05122__	1200	J, K	BP	E
M49467_05122__	1200	K, M	BR, BZ	E
M49467P05152__	1500	J, K	BP	F
M49467_05152__	1500	K, M	BR, BZ	E
M49467P05182__	1800	J, K	BP	F
M49467_05182__	1800	K, M	BR, BZ	E
M49467P05222__	2200	J, K	BP	F
M49467_05222__	2200	K, M	BR, BZ	E
M49467P05272__	2700	J, K	BP	G
M49467R05272__	2700	K, M	BR	F
M49467Z05272__	2700	K, M	BZ	E
M49467P05332__	3300	J, K	BP	G
M49467R05332__	3300	K, M	BR	F
M49467Z05332__	3300	K, M	BZ	E
M49467R05392__	3900	K, M	BR	F
M49467Z05392__	3900	K, M	BZ	E
M49467R05472__	4700	K, M	BR	F
M49467Z05472__	4700	K, M	BZ	E

See footnote at end of table.

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TABLE I. Capacitor characteristics - Continued.

PIN 1/	Capacitance	Capacitance tolerance	Characteristic	Case code
M49467R05562__	5600 pF	K, M	BR	F
M49467Z05562__	5600	K, M	BZ	E
M49467R05682__	6800	K, M	BR	G
M49467Z05682__	6800	K, M	BZ	E
M49467R05822__	8200	K, M	BR	G
M49467Z05822__	8200	K, M	BZ	F
M49467R05103__	.01 µF	K, M	BR	J
M49467Z05103__	.01	K, M	BZ	F
M49467R05123__	.012	K, M	BR	J
M49467Z05123__	.012	K, M	BZ	G
M49467R05153__	.015	K, M	BR	J
M49467Z05153__	.015	K, M	BZ	G
M49467R05183__	.018	K, M	BR	K
M49467Z05183__	.018	K, M	BZ	J
M49467R05223__	.022	K, M	BR	K
M49467Z05223__	.022	K, M	BZ	J
M49467Z05273__	.027	K, M	BZ	K
M49467Z05333__	.033	K, M	BZ	K

1/ The complete PIN shall include additional letter symbols to indicate the characteristic, where applicable, the capacitance tolerance, and the failure rate level.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - MI
Air Force - 11, 19
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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